

Wrangling and Preparing PIT Tag Data using PITcleanr

Kevin See

WDFW
Department of Fish Science

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Washington
Department of
FISH and
WILDLIFE



Today's Journey

- Benefits and challenges of PIT tag data

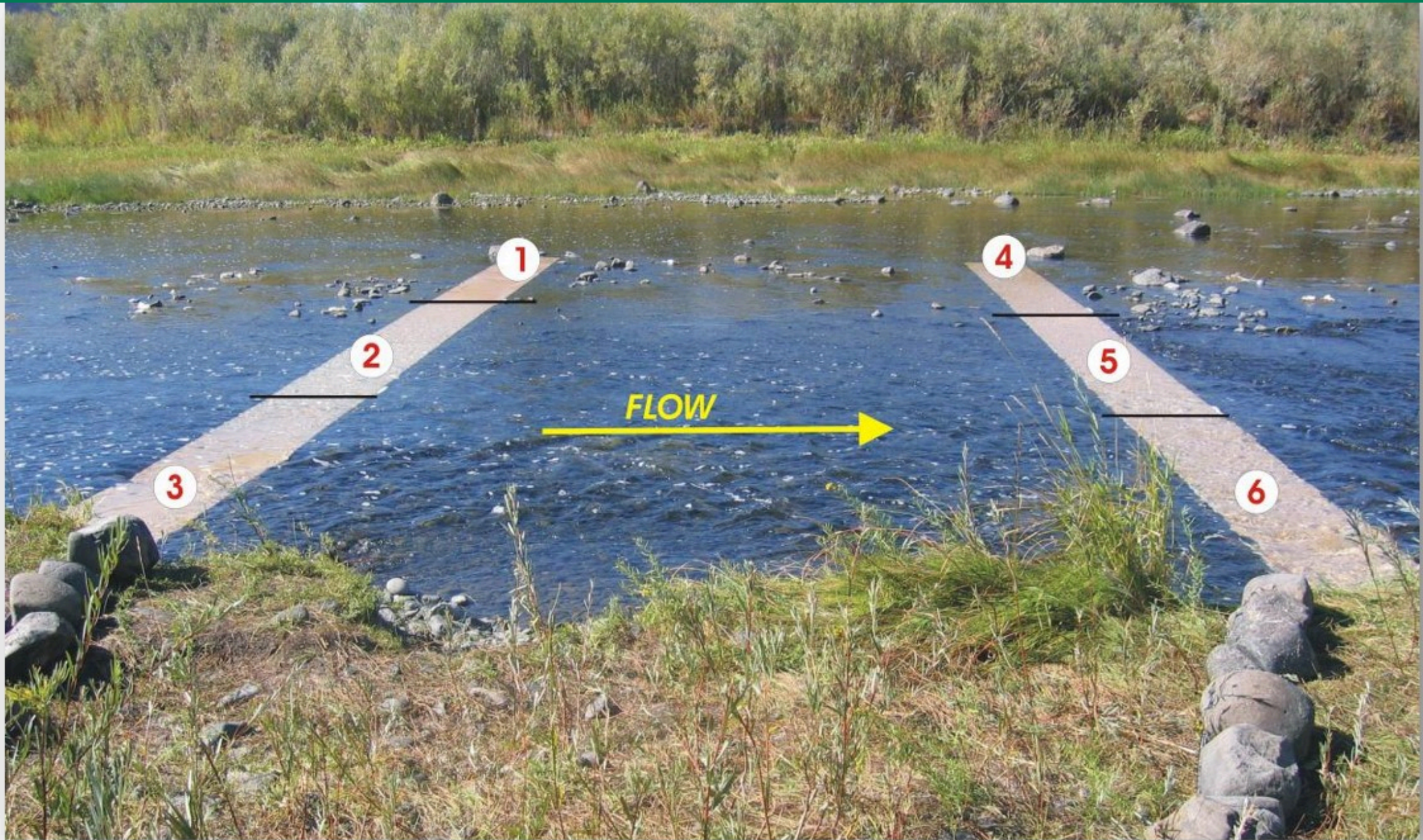
- PITcleanr:

<https://github.com/kevinsee/PITcleanr/>

- Purpose
 - Functionality
 - Package resources
- Summary



PIT Tag Data



PIT Tag Data

Benefits

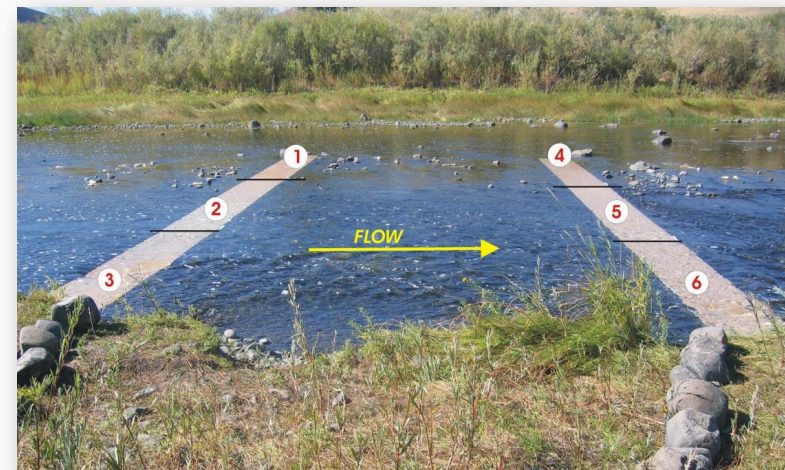
Information about individual fish

- movement
- timing
- survival



Challenges

- **TONS** of data
- Can be overwhelming
- E.g. One juvenile fish detected **681** times at single site within 12 hour period on Lemhi River



So much data...

```
# the complete tag history file from PTAGIS
ptagis_file <-
  system.file("extdata",
              "TUM_chnk_cth_2018.csv",
              package = "PITcleanr")

# read the complete tag history file into R
observations <-
  readCTH(ptagis_file,
          file_type = "PTAGIS")

n_distinct(observations$tag_code) # unique tag codes
```

```
## [1] 1406
```

```
n_distinct(observations$event_site_code_value) # unique event sites
```

```
## [1] 50
```

```
nrow(observations) # total observations
```

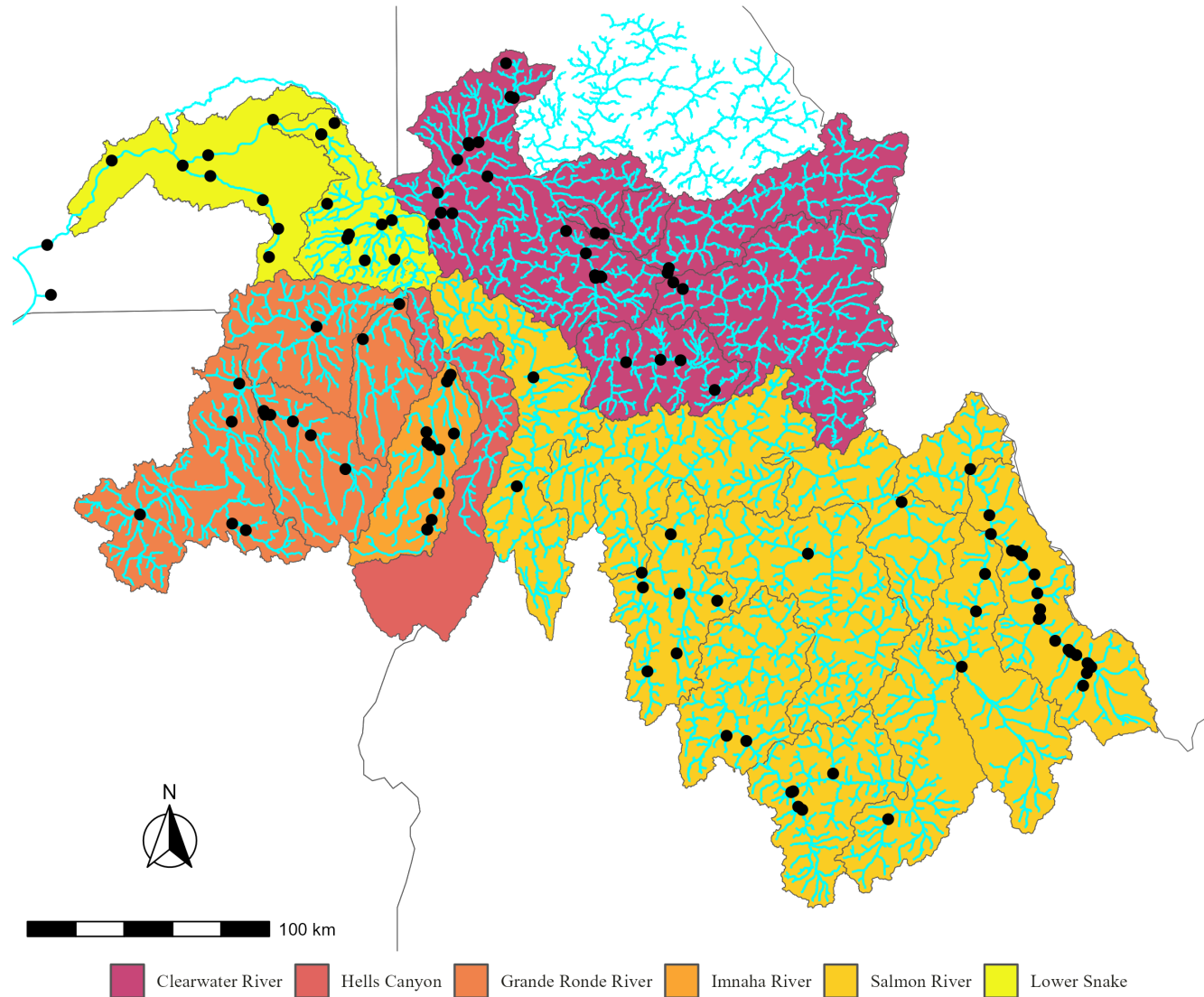
```
## [1] 13501
```


So much data...

Search:

Tag Code	Event Site Code Value	Event Date Time Value	Antenna Id	Antenna Group Configuration Value	Mark Species Name	Mark Rear Type Name	Event Type Name	Event Site Type Description	Event Release Site Code Code	Event Release Date Time Value	Cth Count
All	All	All	All	All	All	All	All	All	All	All	All
1	384.3B239AC47B	NASONC	2016-03-01T10:34:00Z	0	Chinook	Wild Fish or Natural Production	Mark	River	NASONC	2016-03-04T17:00:00Z	1
2	384.3B239AC47B	BO3	2018-05-18T09:36:59Z	18	110	Chinook	Wild Fish or Natural Production	Observation	Adult Fishway		1
3	384.3B239AC47B	BO3	2018-05-18T09:37:11Z	16	110	Chinook	Wild Fish or Natural Production	Observation	Adult Fishway		1
4	384.3B239AC47B	BO3	2018-05-18T09:37:12Z	16	110	Chinook	Wild Fish or Natural Production	Observation	Adult Fishway		1
5	384.3B239AC47B	BO3	2018-05-18T09:42:16Z	14	110	Chinook	Wild Fish or Natural Production	Observation	Adult Fishway		1
2018-05-18T09:42:16Z						Wild Fish or					

Showing 1 to 100 of 100 entries



PITcleanr Package

PITcleanr Purpose

Make your life easier

- Uses 
- Query some data from PTAGIS
 - Metadata about sites
 - Configuration files
 - MRR files
- Read in data from PTAGIS or non-PTAGIS sources
 - Consistent format
 - Identify orphan or disowned tags
- Compress observations
 - Reduce redundant detections
- Determine movement direction
 - Based on parent-child table
 - Includes tools to build parent-child table
- Estimate array/site efficiency
- Build capture history matrix

PITcleanr Queries

PTAGIS

Site Metadata

- `queryPtagisMeta()` - Metadata for all PTAGIS sites
 - `queryInterrogationMeta()`
 - `queryInterrogationConfig()`
 - `queryMRRMeta()`

Tag Details

- `queryMRRDataFile()`
- `queryTimerTagSite()` - can be slow
- `queryCapHist()` - requires API key
- `queryTagEvents()` - requires API key

Non-PTAGIS

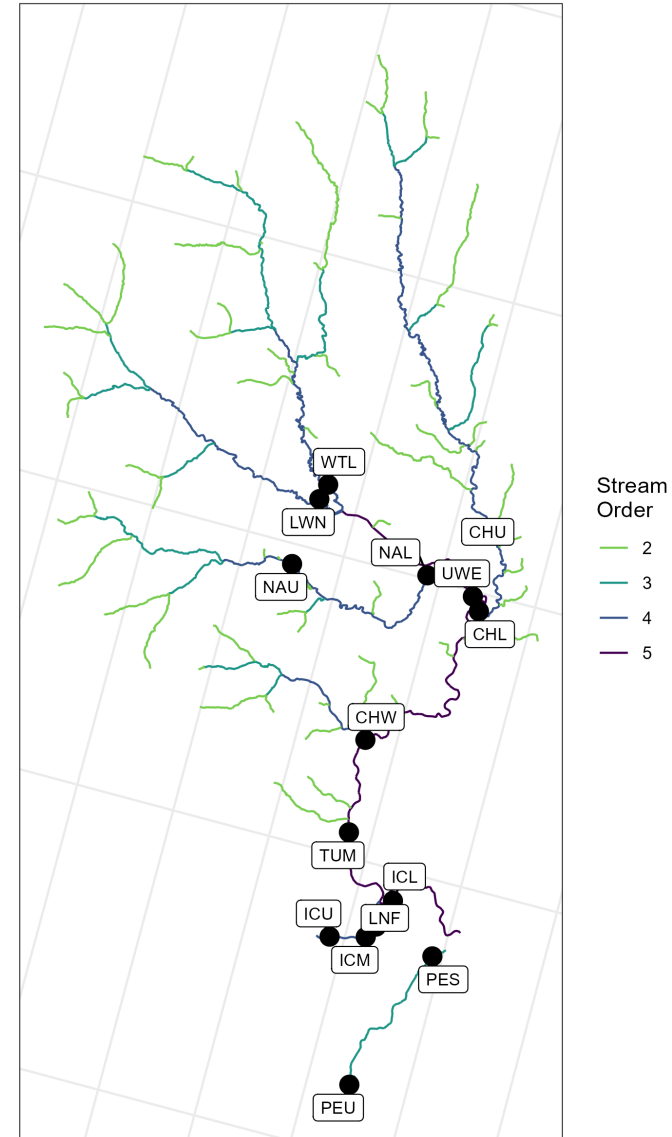
NHDPlus Flowlines

- `queryFlowlines()`
 - uses `nhdplusTools` package from USGS

Example Data

Adult Spring Chinook

- Tagged at Tumwater Dam on Wenatchee River as adults
- Goal is to estimate movement or survival past each site
 - Abundance estimates
 - Accounts for imperfect detection efficiency



Read in Data

Download complete tag histories from PTAGIS as csv file

- Start with a list of tags
- Need certain columns
 - Tag, Event Site Code, Event Date Time, Antenna, Antenna Group Configuration

The screenshot displays the PTAGIS BI 'PITcleanr Query' interface. On the left, an 'INDEX' sidebar lists 18 items, with '1 Select Attributes (Required)' and '2 Select Metrics (Required)' marked as completed. The main area is titled '1. Select Attributes (Required)' and prompts the user to 'Select the attributes to include on your report:'. Below this is a search bar and a list of 'Available' attributes. On the right, a 'Selected:' list shows the chosen attributes: Tag, Event Site Code, Event Date Time, Antenna, and Antenna Group Configuration. At the bottom, there is a 'Report Message Name' field containing 'PITcleanr Query' and navigation buttons: '< Previous', 'Next >', 'Run Report', and 'Cancel'.

PTAGIS BI > My Reports > PITcleanr Query

INDEX

- Summary of your selections
- 1 Select Attributes (Required) ✓
- 2 Select Metrics (Required) ✓
- 3 Event Data Project (used to be Coordinator)
- 4 Event Date
- 5 Event File
- 6 Event Site
- 7 Event Site RKM
- 8 Event Type
- 9 Event Year
- 10 Mark Data Project (used to be Coordinator)
- 11 Mark Date
- 12 Mark File
- 13 Mark File List
- 14 Mark Site
- 15 Mark Site Subbasin
- 16 Mark Year
- 17 Release Date
- 18 Release Site

1. Select Attributes (Required)

Select the attributes to include on your report:
This prompt requires at least one selection.

Search for:

Available:

- Mark Site
- Mark Site Basin
- Mark Site Code
- Mark Site Info
- Mark Site Latitude
- Mark Site Longitude
- Mark Site RKM
- Mark Site Subbasin
- Mark Site Type
- Mark Week
- Mark Year
- Migration Year
- Obs Date Time
- Radio Tag
- Release Date

Selected:

- Tag
- Event Site Code
- Event Date Time
- Antenna
- Antenna Group Configuration

Report Message Name: PITcleanr Query

< Previous Next > Run Report Cancel

Read in Data

Download complete tag histories from PTAGIS as csv file

- Start with a list of tags
- Need certain columns
 - Tag, Event Site Code, Event Date Time, Antenna, Antenna Group Configuration

```
print(cth_file_nm)
```

```
## [1] "TUM_chnk_cth_2018.csv"
```

Use `readCTH()` function:

```
# read the complete tag history file into R
cth_df <-
  system.file("extdata",
              cth_file_nm,
              package = "PITcleanr") >
  readCTH(file_type = "PTAGIS")
```

- PTAGIS not the only format
 - Biologic
 - Raw .log, .xlsx, or .txt files
- Change `file_type`

Compress data

Key feature of PITcleanr

Map each site / antenna to a user-defined *node*

Start with:

- 13,501 observations of
- 1,406 unique tags from
- 50 different sites.

```
comp_obs ← compress(cth_df)
```

End with 4,435 rows of data.

- 32.8% of original data

Compress data

Search:

	Tag Code	Node	Slot	Event Type Name	N Dets	Min Det	Max Det	Duration	Travel Time
	All	All	All	All	All	All	All	All	All
1	384.3B239AC47B	NASONC	1	Mark	1	2016-03-04T17:00:00Z	2016-03-04T17:00:00Z	0	
2	384.3B239AC47B	BO3	2	Observation	18	2018-05-18T09:36:59Z	2018-05-18T12:00:05Z	143.1	1158757
3	384.3B239AC47B	BO4	3	Observation	7	2018-05-18T13:03:11Z	2018-05-18T13:09:29Z	6.3	63.1
4	384.3B239AC47B	TD1	4	Observation	4	2018-05-20T17:20:39Z	2018-05-20T17:20:44Z	0.1	3131.2
5	384.3B239AC47B	JO1	5	Observation	2	2018-05-21T17:49:21Z	2018-05-21T17:51:20Z	2	1468.6
6	384.3B239AC47B	MC2	6	Observation	12	2018-05-24T09:04:38Z	2018-05-24T09:54:53Z	50.2	3793.3

Showing 1 to 100 of 100 entries

Compress data

1. Shrink size of data
2. Assign (or map) observations to node of your choice
3. First, last and number of detections on each node
4. Total duration between first and last detections
5. Travel time from previous node

Configuration File

Configuration file

Map each detection to a user-defined node

Node could be:

- Individual antenna
- Array (group of antennas)
- Site
- Group of sites

Can use PTAGIS site metadata to construct

```
my_config ←  
buildConfig(node_assign = "site")
```

Configuration file

Essential columns

Search:

	Site Code	Antenna Id	Config Id	Node
	All	All	All	All
1	0HR	01	100	0HR
2	0HR	02	100	0HR
3	0HR	03	100	0HR
4	158	D1	100	158
5	158	D2	100	158
6	158	D3	100	158
7	158	D4	100	158
8	158	D5	100	158
9	158	D6	100	158

Showing 1 to 200 of 200 entries

Configuration file

But PTAGIS columns include:

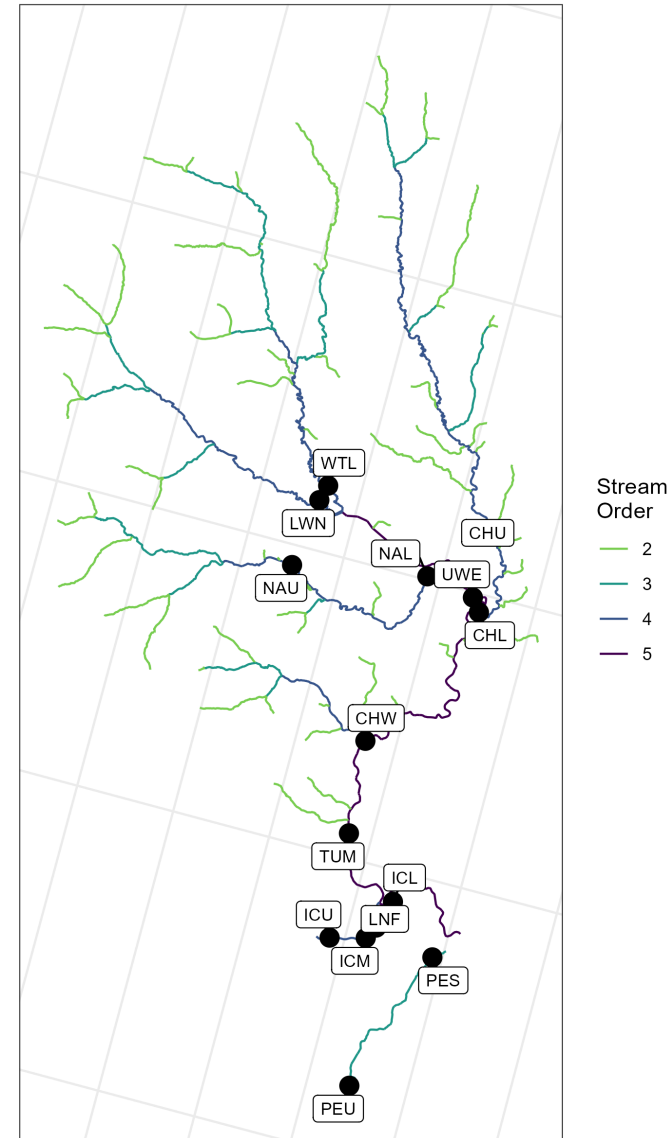
```
names(my_config) ▷  
janitor::make_clean_names("title")
```

##	[1]	"Site Code"	"Config Id"	"Antenna Id"	"Node"
##	[5]	"Start Date"	"End Date"	"Site Type"	"Site Name"
##	[9]	"Antenna Group"	"Site Description"	"Site Type Name"	"Rkm"
##	[13]	"Rkm Total"	"Latitude"	"Longitude"	

Directionality

Map of Sites

- Which sites are upstream / downstream of one another?
- What are possible paths a fish might take?



Parent-Child Table

- Tags move from parent sites to child sites
- Default is to tell which sites are upstream of a particular site
- Can be reversed for downstream movement
- Can be made by hand...
- PITcleanr has tools to help

```
parent_child ▷  
  select(parent, child)
```

```
## # A tibble: 14 × 2  
##   parent child  
##   <chr>   <chr>  
## 1 ICL     LNF  
## 2 ICL     ICM  
## 3 TUM     ICL  
## 4 TUM     UWE  
## 5 TUM     CHL  
## 6 TUM     PES  
## 7 TUM     CHW  
## 8 ICM     ICU  
## 9 UWE     LWN  
## 10 UWE    WTL  
## 11 UWE    NAL  
## 12 CHL    CHU  
## 13 PES    PEU  
## 14 NAL    NAU
```


Parent-Child Table

- PITcleanr has a `buildParentChild()` function.
- Requires an `sf` point object of sites
 - Build yourself or use `extractSites()`
- Requires an `sf` line object of the NHDPlus flowlines, downloaded from USGS.
 - Use `queryFlowlines()` to download it

Build sf object of sites

One way

```
# extract sites from detection file
sites_sf <-
  extractSites(cth_file = cth_df,
               as_sf = T,
               # drop juvenile detection/markings sites
               min_date = "20180501")

# focus on sites within Wenatchee subbasin, and drop a few sites we don't care about
sites_sf <-
  sites_sf %>%
  # all sites in the Wenatchee have a river kilometer that starts with 754
  filter(str_detect(rkm, "^754."),
         type != "MRR",
         site_code != "LWE") >
  mutate(across(site_code,
                ~ recode(.,
                        "TUF" = "TUM"))))
```

Build sf object of sites

Another way

```
# type site codes by hand
sites_sf <-
  tibble(site_code =
    c("ICL", "TUM", "ICM", "UWE", "CHL", "PES", "NAL",
      "LNF", "CHW", "ICU", "LWN", "WTL", "CHU", "PEU",
      "NAU")) >
  left_join(my_config >
    select(site_code,
           site_name,
           site_type_name,
           site_type,
           rkm,
           site_description,
           latitude,
           longitude) >
    distinct(),
    by = join_by(site_code)) >
  st_as_sf(coords = c("longitude",
                     "latitude"),
           crs = 4326) >
  arrange(rkm)
```

Query Flowlines

```
# query the flowlines
nhd_list ←
  queryFlowlines(sites_sf = sites_sf,
                 root_site_code = "TUM",
                 min_strm_order = 2)

# extract flowlines from nhd_list object
flowlines ←
  nhd_list$flowlines
```

```
flowlines ▷
  st_drop_geometry() ▷
  select(comid,
         hydroseq,
         uphydroseq,
         dnhydroseq)
```

```
## # A tibble: 393 × 4
##       comid hydroseq uphydroseq dnhydroseq
##       <int>    <dbl>      <dbl>      <dbl>
## 1 23081168 50054839    50058960    50051361
## 2 23082362 50137692    50220300    50089555
## 3 23081164 50048405    50051361    50045832
## 4 23081166 50051361    50054839    50048405
## 5 23081782 50070209    50078367    50063922
## 6 23081786 50089552    50106443    50078367
## 7 23081158 50137768    50220562    50106470
## 8 23081784 50078367    50089552    50070209
## 9 23081176 50089577    50106469    50078388
## 10 23080678 50014187    50014320    50014062
## # i 383 more rows
```

Build Parent-Child Table

```
parent_child ←  
  suppressWarnings(  
    buildParentChild(sites_sf,  
                     flowlines))
```

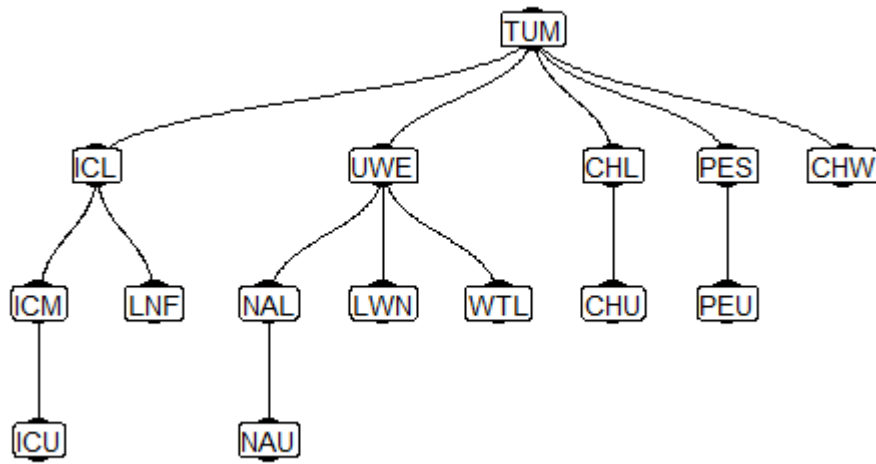
```
parent_child ←  
  editParentChild(parent_child,  
    fix_list =  
      list(c(NA, "PES", "TUM"),  
           c(NA, "LNF", "ICL"),  
           c(NA, "ICL", "TUM"),  
           c("PES", "ICL", "TUM")),  
    switch_parent_child =  
      list(c("ICL", "TUM")))
```

```
# view corrected parent_child table  
parent_child ▶  
  select(parent, child)
```

```
## # A tibble: 14 × 2  
##   parent child  
##   <chr> <chr>  
## 1 ICL    LNF  
## 2 ICL    ICM  
## 3 TUM    ICL  
## 4 TUM    UWE  
## 5 TUM    CHL  
## 6 TUM    PES  
## 7 TUM    CHW  
## 8 ICM    ICU  
## 9 UWE    LWN  
## 10 UWE   WTL  
## 11 UWE   NAL  
## 12 CHL   CHU  
## 13 PES   PEU  
## 14 NAL   NAU
```

Plot Parent-Child Table

```
plotNodes(parent_child,  
  point_size = 10,  
  label_size = 5)
```



All possible fish pathways

```
buildNodeOrder(parent_child)
```

```
## # A tibble: 15 × 3  
##   node node_order path  
##   <chr>      <dbl> <chr>  
## 1 TUM                1 TUM  
## 2 CHL                2 TUM CHL  
## 3 CHU                3 TUM CHL CHU  
## 4 CHW                2 TUM CHW  
## 5 ICL                2 TUM ICL  
## 6 ICM                3 TUM ICL ICM  
## 7 ICU                4 TUM ICL ICM ICU  
## 8 LNF                3 TUM ICL LNF  
## 9 PES                2 TUM PES  
## 10 PEU               3 TUM PES PEU  
## 11 UWE               2 TUM UWE  
## 12 LWN               3 TUM UWE LWN  
## 13 NAL               3 TUM UWE NAL  
## 14 NAU               4 TUM UWE NAL NAU  
## 15 WTL               3 TUM UWE WTL
```

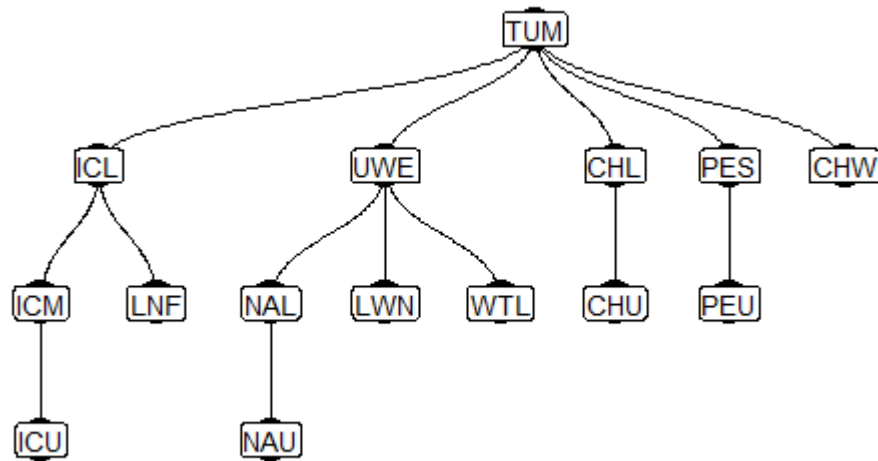
Add Direction

```
comp_dir ←  
  addDirection(compress_obs = comp_obs,  
               parent_child = parent_child)
```

```
comp_dir ▷  
  select(tag_code,  
         node,  
         min_det,  
         node_order,  
         path,  
         direction)
```

```
## # A tibble: 3,257 × 6  
##   tag_code      node min_det      node_order path      direction  
##   <chr>      <chr> <dtm>      <dbl> <chr>      <chr>  
## 1 384.3B239AC47B TUM   2018-06-16 15:26:31      1 TUM        start  
## 2 384.3B239AC47B UWE   2018-06-23 22:03:13      2 TUM UWE      forward  
## 3 384.3B239AC47B NAL   2018-07-01 00:06:16      3 TUM UWE NAL      forward  
## 4 3DD.003B9DD720 TUM   2018-06-17 09:39:26      1 TUM        start  
## 5 3DD.003B9DD725 TUM   2018-06-19 09:33:08      1 TUM        start  
## 6 3DD.003B9DD725 CHU   2018-06-28 21:34:58      3 TUM CHL CHU      forward  
## 7 3DD.003B9DDAB0 TUM   2018-07-17 11:51:44      1 TUM        start  
## 8 3DD.003B9F123C TUM   2018-06-21 09:45:14      1 TUM        start  
## 9 3DD.003B9F123C UWE   2018-07-01 12:55:31      2 TUM UWE      forward  
## 10 3DD.003BC7A654 TUM   2018-06-09 22:36:03      1 TUM        start  
## # i 3,247 more rows
```

Strange Directions



A tibble: 8 × 6

##	tag_code	node	min_det	node_order	path	direction
##	<chr>	<chr>	<dtm>	<dbl>	<chr>	<chr>
## 1	3DD.003BC80D81	TUM	2018-06-21 16:14:26	1	TUM	start
## 2	3DD.003BC80D81	UWE	2018-06-29 15:29:19	2	TUM UWE	forward
## 3	3DD.003BC80D81	NAL	2018-06-30 01:35:27	3	TUM UWE NAL	forward
## 4	3DD.003BC80D81	LWN	2018-07-05 23:05:02	3	TUM UWE LWN	unknown
## 5	3DD.003BC80D81	NAL	2018-07-08 03:06:42	3	TUM UWE NAL	unknown
## 6	3DD.003BC80D81	UWE	2018-07-17 21:52:42	2	TUM UWE	backward
## 7	3DD.003BC80D81	NAL	2018-07-20 21:50:13	3	TUM UWE NAL	forward
## 8	3DD.003BC80D81	NAU	2018-07-30 22:40:06	4	TUM UWE NAL NAU	forward

Filter Observations

```
comp_filter <-  
  filterDetections(compress_obs = comp_obs,  
                  parent_child = parent_child,  
                  max_obs_date = "20180930")
```

```
comp_filter >  
  filter(tag_code %in% examp_tag) >  
  select(tag_code, node, min_det, direction,  
         path,  
         ends_with("keep_obs"))
```

```
## # A tibble: 8 × 7
```

##	tag_code	node	min_det	direction	path	auto_keep_obs	user_keep_obs
##	<chr>	<chr>	<dtm>	<chr>	<chr>	<lgl>	<lgl>
## 1	3DD.003...	TUM	2018-06-21 16:14:26	start	TUM	TRUE	NA
## 2	3DD.003...	UWE	2018-06-29 15:29:19	forward	TUM ...	FALSE	NA
## 3	3DD.003...	NAL	2018-06-30 01:35:27	forward	TUM ...	FALSE	NA
## 4	3DD.003...	LWN	2018-07-05 23:05:02	unknown	TUM ...	FALSE	NA
## 5	3DD.003...	NAL	2018-07-08 03:06:42	unknown	TUM ...	FALSE	NA
## 6	3DD.003...	UWE	2018-07-17 21:52:42	backward	TUM ...	TRUE	NA
## 7	3DD.003...	NAL	2018-07-20 21:50:13	forward	TUM ...	TRUE	NA
## 8	3DD.003...	NAU	2018-07-30 22:40:06	forward	TUM ...	TRUE	NA

Other Functionality

Array Efficiency

Based on tags detected at an array, and any tags detected upstream of that array

Start by building a description of how each node is connected to others

```
node_order ←  
  buildNodeOrder(parent_child)
```

```
node_order
```

```
## # A tibble: 15 × 3  
##   node node_order path  
##   <chr>      <dbl> <chr>  
## 1 TUM          1 TUM  
## 2 CHL          2 TUM CHL  
## 3 CHU          3 TUM CHL CHU  
## 4 CHW          2 TUM CHW  
## 5 ICL          2 TUM ICL  
## 6 ICM          3 TUM ICL ICM  
## 7 ICU          4 TUM ICL ICM ICU  
## 8 LNF          3 TUM ICL LNF  
## 9 PES          2 TUM PES  
## 10 PEU         3 TUM PES PEU  
## 11 UWE          2 TUM UWE  
## 12 LWN          3 TUM UWE LWN  
## 13 NAL          3 TUM UWE NAL  
## 14 NAU          4 TUM UWE NAL NAU  
## 15 WTL          3 TUM UWE WTL
```

Array Efficiency

Based on tags detected at an array, and any tags detected upstream of that array

```
estNodeEff(comp_obs,  
            node_order)
```

If no node is supplied, calculated for all nodes

Search:

	node 	tags_at_node 	tags_above_node 	tags_resighted 	est_tags_at_node 	est_tags_se 	eff_est 	eff_se 
	All	All	All	All	All	All	All	All
1	ICL	6	20	1	72	34.1	0.083	0.039
2	CHL	441	454	165	1211	59	0.364	0.018
3	NAL	168	133	83	269	12.7	0.625	0.029
4	UWE	361	256	175	528	16	0.684	0.021
5	TUM	1406	1039	1039	1406	0	1	0
6	CHU	454	0	0	454	0	1	0
7	CHW	22	0	0	22	0	1	0

Capture Histories

May need matrix of 1s and 0s

```
buildCapHist(comp_filter,  
              parent_child = parent_child,  
              configuration = my_config)
```

```
## # A tibble: 1,406 × 2  
##   tag_code      cap_hist  
##   <chr>      <chr>  
## 1 384.3B239AC47B 100000000010100  
## 2 3DD.003B9DD720 1000000000000000  
## 3 3DD.003B9DD725 1010000000000000  
## 4 3DD.003B9DDAB0 1000000000000000  
## 5 3DD.003B9F123C 100000000010000  
## 6 3DD.003BC7A654 100000000010110  
## 7 3DD.003BC7F0E3 100000000010110  
## 8 3DD.003BC80A3C 100000000010100  
## 9 3DD.003BC80D81 100000000010110  
## 10 3DD.003BC80F20 100000000000100  
## # i 1,396 more rows
```

What do columns stand for?

```
defineCapHistCols(parent_child = parent_child,  
                   configuration = my_config) ▸  
as_tibble()
```

```
## # A tibble: 15 × 1  
##   value  
##   <chr>  
## 1 TUM  
## 2 CHL  
## 3 CHU  
## 4 CHW  
## 5 ICL  
## 6 ICM  
## 7 ICU  
## 8 LNF  
## 9 PES  
## 10 PEU  
## 11 UWE  
## 12 LWN  
## 13 NAL  
## 14 NAU  
## 15 WTL
```

Package Resources

Package Website

- <https://kevinsee.github.io/PITcleanr/>

PITcleanr 3.0.4FUNCTIONSVIGNETTES ▾NEWS

PITcleanr

Description

`PITcleanr` is an R package for preparing PIT tag data for further analysis. The package can help import complete tag histories from `PTAGIS`, build a configuration file to help assign each detection to a "node", and compress those detections into a smaller file. It contains functions to determine which detection locations are upstream or downstream of each other, build a parent-child relationship table describing whether detection locations are upstream or downstream in relation to each other, and assign directionality of movement between each detection location. For analyses that focus on one-way directional movement (e.g., straightforward CJS models), `PITcleanr` can help determine which detections fail to meet that one-way movement assumption and should be examined more closely, and which detections can be kept.

It was originally conceived as a companion to the `DABOM` package for estimating abundance of returning anadromous adult fish moving upstream. `PITcleanr` was designed to prepare the raw PIT tag observations from `PTAGIS` for use in the `DABOM` package.

The user can find more information related to installation and use of this package on the [package website](https://kevinsee.github.io/PITcleanr/), <https://kevinsee.github.io/PITcleanr/>.

Installation Instructions

The `PITcleanr` package can be installed as an R package from GitHub by using Hadley Wickham's `devtools` package:



Links

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Community

[Contributing guide](#)

Citation

[Citing PITcleanr](#)

Developers

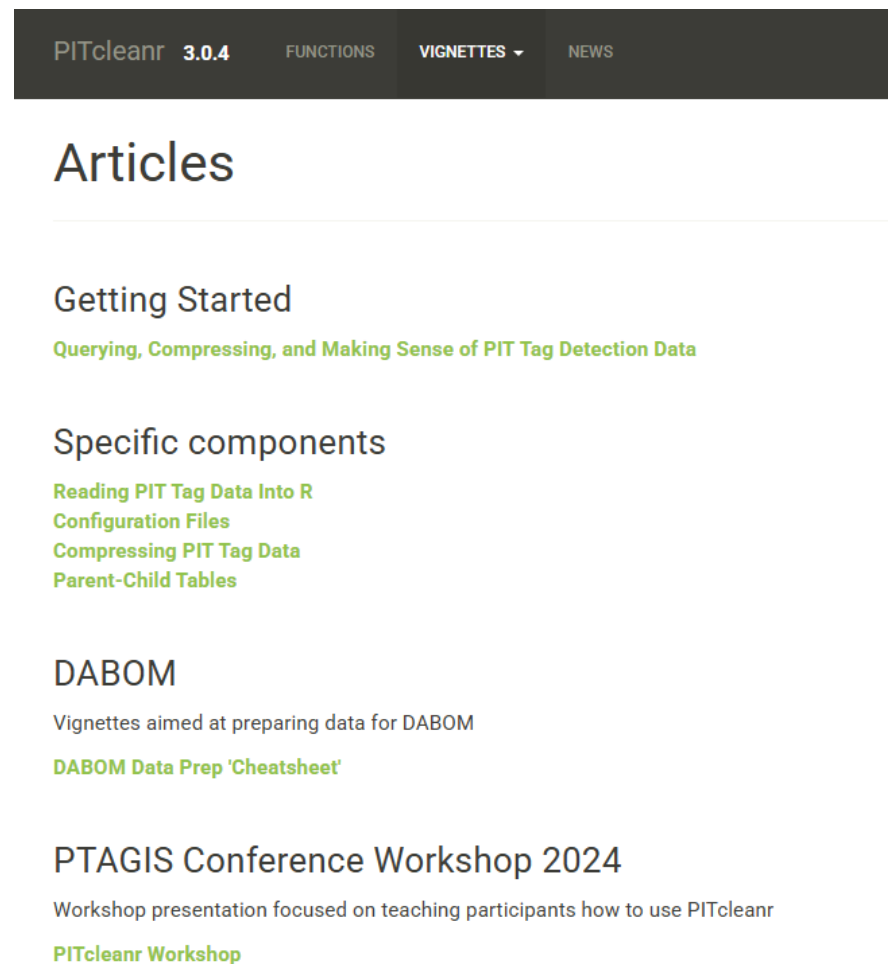
Kevin See
Author, maintainer

Dev status

 launch  binder

Package Website

- <https://kevinsee.github.io/PITcleanr/>



The screenshot shows the top navigation bar of the PITcleanr website. The bar is dark grey with white text. It includes the package name 'PITcleanr' followed by the version '3.0.4'. To the right are four menu items: 'FUNCTIONS', 'VIGNETTES' (with a dropdown arrow), and 'NEWS'. Below the navigation bar, the 'Articles' section is visible, featuring a horizontal line separator. The articles listed are: 'Getting Started' with a sub-link 'Querying, Compressing, and Making Sense of PIT Tag Detection Data'; 'Specific components' with sub-links 'Reading PIT Tag Data Into R', 'Configuration Files', 'Compressing PIT Tag Data', and 'Parent-Child Tables'; 'DABOM' with a sub-link 'Vignettes aimed at preparing data for DABOM' and another sub-link 'DABOM Data Prep 'Cheatsheet''; and 'PTAGIS Conference Workshop 2024' with a sub-link 'Workshop presentation focused on teaching participants how to use PITcleanr' and another sub-link 'PITcleanr Workshop'.

PITcleanr 3.0.4 FUNCTIONS VIGNETTES ▾ NEWS

Articles

Getting Started

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Specific components

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[Compressing PIT Tag Data](#)
[Parent-Child Tables](#)

DABOM

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Contact Developers

- **Kevin See:** Kevin.See@dfw.wa.gov
- **Mike Ackerman:** MikeA@nezperce.org
- **Ryan Kinzer:** RyanK@nezperce.org



Questions?